THE CRISPRO YC PLAYBOOK

A-to-Z Application & Demo Guide

Matching cancer patients to trials where their biology will actually respond

CrisPRO.ai

CONTENTS

PART 1: The Problem -- Why Oncology Trials Fail

PART 2: The One-Liner (Plain English First)

PART 3: The Founder Story

PART 4: The Traction Section

PART 5: The Market

PART 6: The Demo (3-Minute Structure)

PART 7: YC Interview Q&A

PART 8: Pre-Submission Checklist + Appendix

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PART 1: THE PROBLEM -- WHY ONCOLOGY TRIALS FAIL

YC Application Field: "What problem are you solving?"

71% of Phase II oncology trials fail. The standard explanation is that drugs do not work. The real explanation is that patients are matched to trials using Boolean eligibility checklists that ignore whether their tumor biology will actually respond to the drug's mechanism of action.

> *"The bottleneck in oncology is not drug efficacy -- it's patient selection. 53.7% of patients who qualify for trials don't have the biology to respond. We can detect that before enrollment."*

The Evidence

Finding	Data	Source
71% Phase II failure rate	Industry-wide oncology trial failure rate	Hay et al. / BIO industry analysis
53.7% target-alignment gap	314/585 ovarian cancer patients met eligibility but lacked mechanism alignment	AACR 2026 LB340 (CrisPRO)
6x ORR difference	Q4 patients: 42.9% ORR vs Q1 patients: 7.1% ORR	PATH A quartile stratification
IMblaze370 false positive	fit=0.856 (high) -- OS HR=1.00 (zero benefit) $300M spent, zero survival benefit	ClinicalTrials.gov / NEJM
AUROC 0.689	Brain metastasis prediction validated in MSK-MET cohort (N=25,775)	AACR 2026 B065 (CrisPRO)

What CrisPRO Does

CrisPRO matches cancer patients to clinical trials based on whether their tumor biology will actually respond -- not just whether they meet the eligibility checklist. The platform integrates CRISPRa functional genomics screens, Evo2 genomic foundation models, and a validated mechanism-aware ranker (PATH A) to confirm biological fit before enrollment.

WHAT CRISPRO IS NOT	WHAT CRISPRO IS
A citation-checking tool	A multimodal AI platform for cancer dependency discovery and mechanism-aware trial matching
A pharma BD document tool	Simultaneously platform developer AND clinical trial sponsor
A hallucination firewall product	A Glass Box architecture replacing Black Box oncology

A variant annotation service	A functional genomics engine with CRISPRa screens across 18,858 genes
A trial search engine	A validated PATH A ranker tested across 8 MSS CRC IO trials

PART 2: THE ONE-LINER

YC Application Field: "What does your company do?"

> *CORRECT ONE-LINER: "CrisPRO matches cancer patients to clinical trials based on whether their tumor biology will actually respond -- not just whether they meet the eligibility checklist. We've validated this in 8 failed trials: patients who qualified on paper but lacked the biology to respond can be identified before enrollment."*

Why This Works for a Non-Specialist YC Partner

The one-liner above leads with the clinical outcome in plain English. A partner who has never heard of CRISPRa or PATH A understands the problem and the solution from the first sentence. The technology belongs in the follow-up sentence when the partner asks "how?"

When They Ask...	Say This
What do you do? (first 10 seconds)	We match cancer patients to trials based on whether their biology will actually respond -- not just whether they
How does it work? (follow-up)	We integrate CRISPRa functional screens across 18,858 genes, Evo2 genomic foundation models, and a PA
What's your evidence?	53.7% of ovarian cancer patients who qualified for trials lacked the biology to respond (N=585, AACR 2026).
What makes you different?	We are simultaneously platform developer and clinical trial sponsor. We run BreAK CRC-001 (NCT06934538)

The Four-Layer Platform Architecture

Layer	Component	Function
1	Genomic Foundation Models	Evo2 (9.3T tokens), Enformer (chromatin), AlphaFold 3 (3D structure) Digital twin for therapeutic design
2	Functional Genomics Engine	CRISPRa screens (GSE237446, 18,858 genes), DepMap 25Q3, GDSC2, ATAC-seq Cancer dependency discovery, synthetic lethality
3	Mechanism-Aware Trial Ranker	PATH A fit = clip((p*t)/||t||2, 0, 1) Validated retrospectively across 8 MSS CRC IO trials
4	Evidence Governance (Glass Box)	VERIFIED / PROVISIONAL / BLOCKED taxonomy Claim admissibility ledger -- no ungrounded inference reaches a clinician

PART 3: THE FOUNDER STORY

YC Application Field: "Please tell us about the founder(s)"

Fahad Kiani -- Founder & CEO

THE HACK

> *"I built a mechanism-aware ranker that retrospectively scored 8 failed oncology trials and correctly flagged every false positive before looking at the outcomes."*

Specifically: IMblaze370 (atezolizumab + cobimetinib, MSS CRC) received a PATH A fit score of 0.856 -- high fit -- yet OS HR = 1.00 (zero benefit). The ranker correctly identified this as a false positive. CO.26 (durvalumab + tremelimumab) emerged as the strongest calibration case with pTMB HR = 0.34. All 8 trials validated retrospectively without outcome data in hand.

THE DETERMINATION

Fahad is running a live Phase I/II clinical trial -- BreAK CRC-001 (NCT06934538) -- in MSS/pMMR colorectal cancer while simultaneously building the platform. This makes him the only founder in this space who is simultaneously platform developer and trial sponsor. Every platform output is stress-tested against a live regulatory environment. Trial amendments (A1, A3) have already been driven by platform findings.

THE NON-OBVIOUS INSIGHT

> *"The bottleneck in oncology is not drug efficacy -- it's patient selection. 53.7% of patients who qualify for trials don't have the biology to respond. We can detect that before enrollment."*

In a cohort of 585 ovarian cancer patients, 314 (53.7%) met standard eligibility criteria but lacked meaningful mechanism alignment with the trial's therapeutic target. This gap is invisible to Boolean eligibility tools. CrisPRO's 7-dimensional mechanism vector framework detects it before a single patient is enrolled.

TECHNICAL DEPTH

* CRISPRa screens: GSE237446, 18,858 genes, brain colonization phenotype
* Evo2: 9.3 trillion-token genomic foundation model for sequence-level fitness scoring
* AlphaFold 3: 3D structural validation of therapeutic targets
* PATH A fit function: clip((p*t)/||t||2, 0, 1) -- validated across 8 MSS CRC IO trials
* MSK-MET validation: N=25,775 patients, 2,921 brain metastasis samples, AUROC 0.689
* DepMap 25Q3 + GDSC2: 900+ cancer cell lines for dependency mapping

The Team

Name	Role	Key Contribution
Fahad Kiani	Founder & CEO Computational AI Engineer	Built rankers, vector versioning, formula governance, validation datasets
Robin Kim	Co-Founder Chief of Medicine (MD, Hospitalist)	Frontline implementation insight, real care settings, clinical plausibility audit

YC FRAMING TIP: Two MD co-founders is a structural advantage -- not just a credential. Every output layer is audited for clinical plausibility before release. This is why CrisPRO can make claims that a pure software company cannot: the clinical review is built into the founding team, not outsourced to an advisory board.

PART 4: THE TRACTION SECTION

YC Application Field: "What is your traction?"

VALIDATED TRACTION

Milestone	Detail	Status
AACR 2026 -- LB340	Mechanism-based trial matching reveals a 54% target alignment gap in ovarian cancer (N=585)	Peer-reviewed Accepted
AACR 2026 -- B065	Stage-aware CRISPR design for brain metastasis interception: Multi-modal validation of BBB extravasation targets	Peer-reviewed Accepted
AACR 2026 -- LB-B013	Eight-Pathway transcriptomic biomarker outperforms PD-L1 for anti-PD-1 response prediction in melanoma	Peer-reviewed Accepted
AACR 2026 -- B025	An agentic platform for designing cancer immunotherapies: From automated variant interpretation to in silico validation	Peer-reviewed Accepted
UC Berkeley AgentX	2nd place, Entrepreneurship Track (rdi.berkeley.edu/agentx)	Awarded
Brenus Pharma	Existing partner -- BreAK CRC-001 (NCT06934538) Live Phase I/II MSS CRC trial	Active
University of Utah Health	Existing partner -- EHR/genomic integration	Active

PIPELINE (IN PROGRESS -- not confirmed partners)

Organization	Relationship	Status
NCI (NIH)	National Cancer Institute -- in contact for collaboration	In Progress
St. Jude Children's Research Hospital	In contact for pediatric oncology collaboration	In Progress
Google.org GIC	$3M Impact Challenge: AI for Science Submitted 05/01/2026	Submitted (Decision Pending)

Key Performance Metrics

VALIDATED METRICS (observed in data)

- 6x ORR difference: Q4 patients 42.9% vs Q1 patients 7.1% (PATH A quartile stratification)
- AUROC 0.689 -- brain metastasis prediction (MSK-MET confirmed, N=25,775 patients, 2,921 brain mets)
- 53.7% target-alignment gap -- ovarian cancer (314/585 patients, AACR 2026 LB340)
- 8 MSS/pMMR CRC IO trials retrospectively validated with PATH A ranker
- IMblaze370 correctly flagged as false positive: fit=0.856, OS HR=1.00 (zero benefit)
- PATH A calibration error: <0.05 fit units against CO.26 anchor (pTMB HR=0.34)

MODEL PROJECTIONS (not observed outcomes -- will be probed in interview)

INTERVIEW PREP -- These are model projections, NOT observed outcomes. Do NOT present alongside validated metrics in the YC application itself. Be prepared to explain the model when asked. | 87.5% projected reduction in Phase 2 oncology failures (Holistic Feasibility Score -- model projection, not observed). | PathwayKELIM AUROC target: 0.78 vs standard KELIM 0.72 (target, not yet achieved).

Datasets Underpinning Validation

Dataset	Description	Scale
GSE237446	CRISPRa screen, brain colonization phenotype	18,858 genes
MSK-MET	Multi-tumor metastasis cohort	25,775 patients; 2,921 brain mets
DepMap 25Q3 + GDSC2	Cancer dependency + drug sensitivity	900+ cell lines
Enformer/Borzoi	Chromatin accessibility predictions	~1M genomic contexts
GSE205033	Brain-metastasis chromatin accessibility (ATAC-seq)	Brain mets samples
ClinicalTrials.gov API v2	MSS/pMMR CRC IO trial records	8 verified trials

PART 5: THE MARKET

YC Application Field: "What is the market size?"

Three-Tier Market Structure

Tier	Market	Size & Rationale
BEACHHEAD (Now)	MSS/pMMR CRC Metastatic IO Programs	~30,000 new metastatic diagnoses/year (US) 20-30 active IO programs globally NO approved immunotherapy -- every sponsor needs mechanism-aware patient selection Each failed Phase II trial costs $50-300M $1-6B annually in failed trial spend in this indication alone (20-30 programs x 71% failure rate x $50-300M per failure)
MID-MARKET (Year 2-3)	All Phase I/II Oncology Programs (US)	~800 active programs Protocol gap engine + claim governance as SaaS Every sponsor faces the same Boolean eligibility problem
LONG-TERM (Year 4+)	Global Precision Oncology	Platform developer + trial sponsor model scales to: NSCLC brain metastasis (already validated) Ovarian cancer (already validated) Metastatic breast cancer Pharma Partnership Program: salvage Phase 2 assets

Revenue Model

Revenue Stream	Description	Target Customer
SaaS Licensing	Protocol gap engine + claim governance Subscription access to mechanism-aware ranker	Oncology trial sponsors
Translational Consulting	Amendment-ready protocol specifications Mechanism vector analysis for existing trials	Clinical-stage biotechs
Milestone-Linked Partnerships	Academic collaborator revenue tied to validation milestones	NCI, St. Jude, U of Utah
Pharma Partnership Program	License in-silico trial framework to salvage Phase 2 assets -- identify responders in previously failed unselected trials	Big pharma (Phase 2 rescue)
Follow-on Philanthropy	Grant funding for open-source platform components and rare disease expansion	Google.org, NIH, foundations

Scale Path

Policy & Standards Leadership: Define the ethical gold standard for mechanism-aware trial matching in oncology

Geographic Transfer: Local pilot (BreAK CRC-001, Utah) -> regional -> global

Technical & Performance Maturity: Enterprise-scale throughput across 56,083 genes at production scale

Community-Led Expansion: Open-source contributor base for platform components

Pharma Partnership Program: License in-silico trial framework to salvage Phase 2 assets

PART 6: THE DEMO -- 3-MINUTE YC DEMO STRUCTURE

How to Demo CrisPRO in 3 Minutes

WHAT NOT TO DEMO: Do NOT demo the Plumber/citation-checking workflow. Do NOT demo the REGISTRY.yaml or blocker ledger. Those are internal governance tools, not the product. Lead with clinical outcomes, not infrastructure.

MINUTE 1 -- The Problem Made Visceral

Step	What to Show	Key Message
1a	IMblaze370 trial card Atezolizumab + cobimetinib, MSS CRC	$300M spent. OS HR = 1.00. Zero benefit. This is what a failed trial looks like.
1b	PATH A ranker output for IMblaze370 fit score = 0.856	High fit score -- standard eligibility tools would have enrolled these patients.
1c	The contradiction: high fit, null outcome	This is the false positive problem. CrisPRO flags it. Standard tools do not.

MINUTE 2 -- The Platform in Action

Step	What to Show	Key Message
2a	7-dimensional mechanism vector for a real ovarian cancer patient	She meets standard eligibility criteria. On paper, she qualifies.
2b	Mechanism vector alignment score 53.7% misalignment with trial MOA	Her biology does not match the trial's mechanism of action. She will not respond.
2c	VERIFIED/PROVISIONAL/BLOCKED claim governance output	Every claim is tagged. No ungrounded inference reaches the clinician.
2d	Protocol gap engine output Amendment language generated	CrisPRO does not just flag the gap -- it generates the fix.

MINUTE 3 -- The Closed Feedback Loop (The Moat)

Step	What to Show	Key Message
3a	BreAK CRC-001 trial page NCT06934538	CrisPRO is not just a software tool. It IS the platform running this trial.
3b	The feedback loop: Platform output -> Amendment A1/A3 -> IRB submission -> Outcome data -> Ranker recalibration	This loop is the moat. Every trial amendment recalibrates the ranker. No pure software company can do this.

Step	What to Show	Key Message
3c	The structural moat statement	"We are simultaneously platform developer and clinical trial sponsor. This closed feedback loop is not replicable."

> *THE MOAT IN ONE SENTENCE: "We are simultaneously platform developer and clinical trial sponsor -- running BreAK CRC-001 (NCT06934538) in MSS/pMMR CRC while validating mechanism-aware trial routing across 8 failed IO trials, 3 cancer indications, and 4 peer-reviewed AACR 2026 abstracts. This closed feedback loop -- platform output -> trial amendment -> outcome data -> ranker recalibration -- is not replicable by a pure software company without a trial, or a pure biotech without a platform."*

PART 7: YC INTERVIEW Q&A -- 10 MOST LIKELY QUESTIONS

Prepared Answers -- Use Exact Language Where Possible

Q1: What does CrisPRO do?

We match cancer patients to clinical trials based on whether their tumor biology will actually respond -- not just whether they meet the eligibility checklist. 71% of Phase II oncology trials fail -- not because drugs don't work, but because patients are matched on eligibility criteria that ignore molecular mechanism fit. We've validated this: 53.7% of ovarian cancer patients who qualified for trials lacked the biology to respond. Our platform integrates CRISPRa functional screens, Evo2 genomic foundation models, and a PATH A ranker validated across 8 MSS CRC IO trials to route patients where biological fit is confirmed before enrollment.

Q2: What's your unfair advantage?

We are simultaneously platform developer and clinical trial sponsor. We're running BreAK CRC-001 (NCT06934538) -- a live Phase I/II MSS CRC trial -- in the exact indication our platform serves. Every gap the engine identifies gets stress-tested against a live regulatory environment. That closed feedback loop -- platform output -> trial amendment -> outcome data -> ranker recalibration -- is not replicable by a pure software company without a trial, or a pure biotech without a platform.

Q3: What's your traction?

4 peer-reviewed AACR 2026 abstracts. 2nd place UC Berkeley AgentX. Two existing partners (Brenus Pharma, University of Utah Health). $3M Google.org GIC submission filed. Brain metastasis: AUROC 0.689 confirmed in MSK-MET (N=25,775). Ovarian cancer: 53.7% target-alignment gap (N=585). MSS CRC: 8 trials retrospectively validated, IMblaze370 correctly flagged as false positive.

Q4: Who are your competitors?

Existing tools solve one slice -- variant annotation (Foundation Medicine), panel reporting (Tempus), or trial search (TrialSpark). None connect metastatic biology, rare vulnerabilities, and mechanism-aware routing in a single auditable framework. We are the only platform that is also a trial sponsor.

Q5: Why now?

Three things converged. First, genomic foundation models (Evo2, 9.3T tokens) became capable enough to score molecular fit at scale. Second, ClinicalTrials.gov API v2 made protocol intelligence programmatically accessible. Third, the field has now accumulated enough failed trials (IMblaze370, AtezoTRIBE, CO.26) to validate retrospectively -- we have ground truth to train against.

Q6: What's the business model?

SaaS licensing for the protocol gap engine and claim governance layer to oncology sponsors. Translational consulting for amendment-ready protocol specifications. Milestone-linked partnership revenue from academic collaborators. Long-term: Pharma Partnership Program licensing our in-silico trial framework to salvage Phase 2 assets by identifying responders in previously failed unselected trials.

Q7: How big is the market?

Beachhead: ~30,000 new metastatic MSS/pMMR CRC diagnoses annually in the US, 20-30 active IO programs, no approved immunotherapy. Each failed Phase II trial costs $50-300M; with 20-30 active programs and a 71% failure rate, that's $1-6B annually in failed trial spend in this indication alone. Mid-market: ~800 active Phase I/II oncology programs in the US. Long-term: global precision oncology across three already-validated indications.

Q8: What's the IP?

Four USPTO provisional applications filed covering 7D Mechanism Vectors, the re-ranking system, and the Evidence Matrix discovery engine. Utility conversion is in progress. We filed provisionals to establish priority dates while the platform matures -- standard practice for early-stage biotech IP.

Q9: Tell me about the team.

Fahad Kiani (Founder/CEO) -- computational AI engineer, built the rankers, manages vector versioning and formula governance. Robin Kim (Co-Founder, Chief of Medicine) -- MD, hospitalist, frontline implementation insight. Two founders, both with deep domain expertise. Every output layer is audited for clinical plausibility before release -- the clinical review is built into the founding team, not outsourced to an advisory board.

Q10: What do you need the money for?

Three things. First, wet-lab validation ($1.2M): synthesize top 40 CRISPR guides from the Assassin Score, run in-vitro cutting efficiency assays, collect the first prospective GzmB/TOX/MHC-I/TLS biopsy panel from BreAK CRC-001 -- the first prospective trafficking and effector dataset in MSS CRC IO. Second, compute ($500K): Evo2, AlphaFold 3, and mechanism-vector trial matching across 56,083 genes at production scale. Third, regulatory and IP ($295K): FDA pre-submission meeting, CLIA-ready companion diagnostic transition, patent utility conversion.

PART 8: PRE-SUBMISSION CHECKLIST

Final Review Before Submitting YC Application

Use this checklist to verify every section of the YC application before submission. Each item maps to a specific field or claim that YC partners will probe.

[ ] **One-Liner**

Does it lead with the clinical outcome in plain English? ("matches patients to trials based on whether their biology will actually respond") -- NOT with technology terms in the first sentence?

Note: Must NOT open with "CRISPRa", "PATH A", or "Boolean eligibility" -- save those for the follow-up

[ ] **Problem Statement**

Does it cite 71% Phase II failure rate AND 53.7% target-alignment gap AND 6x ORR difference?

Note: All three metrics must appear together

[ ] **Traction**

Does it list all 4 AACR 2026 abstracts (LB340, B065, LB-B013, B025), Berkeley AgentX, AND both existing partners (Brenus Pharma, University of Utah Health)?

Note: NCI and St. Jude are pipeline -- do not present as confirmed partners

[ ] **Team**

Does it explain why two MD co-founders is a structural advantage -- specifically that every output layer is audited for clinical plausibility before release?

Note: Not just credentials -- structural function. Do not list unnamed co-founders.

[ ] **Moat**

Does it explain the closed feedback loop: platform developer + trial sponsor, platform output -> trial amendment -> outcome data -> ranker recalibration?

Note: This is the core differentiation

[ ] **Demo**

Does it show: (1) IMblaze370 false positive detection, (2) mechanism vector misalignment for ovarian cancer patient, (3) BreAK CRC-001 feedback loop?

Note: All three demo beats required

[ ] **Market**

Does it include a dollar figure for the beachhead? ($1-6B annually in failed trial spend in MSS CRC IO alone)

Note: Patient counts alone are not market sizing -- YC partners think in dollars

[ ] **IP**

Does it say "four USPTO provisional applications filed, utility conversion in progress" -- NOT "four-patent portfolio"?

Note: Provisionals are not patents. YC partners know the difference.

[ ] **Metrics**

Are validated metrics (AUROC 0.689, 53.7%, 6x ORR) clearly separated from model projections (87.5% Holistic Feasibility Score)?

Note: Never present a projected metric alongside validated metrics without a clear label

[ ] **Language Audit**

Have you removed all instances of: "AI-powered", "citation-checking tool", "pharma BD document tool", "hallucination firewall"?

Note: These describe components, not the platform

APPENDIX: Internal Positioning Reference

The following is an internal reference only -- do not include in the YC application itself. It documents what CrisPRO is not, to prevent positioning drift.

WHAT CRISPRO IS NOT	WHAT CRISPRO IS
A citation-checking tool	A multimodal AI platform for cancer dependency discovery and mechanism-aware trial matching
A pharma BD document tool	Simultaneously platform developer AND clinical trial sponsor
A hallucination firewall product	A Glass Box architecture replacing Black Box oncology
A variant annotation service	A functional genomics engine with CRISPRa screens across 18,858 genes
A trial search engine	A validated PATH A ranker tested across 8 MSS CRC IO trials

> *"The product is mechanism-aware trial routing. The Glass Box governance layer is an architectural property of the platform -- not the product itself. Never lead with governance; lead with the clinical outcome: patients matched to trials where biological fit is confirmed before enrollment."*

Three Validated Indications -- Quick Reference

Indication	Key Metric	Dataset	Venue
Brain Metastasis	AUROC 0.689 29-gene validation panel	MSK-MET: N=25,775 2,921 brain mets	AACR 2026 B065
Ovarian Cancer	53.7% target-alignment gap (314/585 patients)	N=585 cohort 7D mechanism vectors	AACR 2026 LB340

Indication	Key Metric	Dataset	Venue
MSS/pMMR CRC	8 trials validated IMblaze370 false positive flagged fit=0.856, OS HR=1.00	ClinicalTrials.gov API v2 8 verified trial records	BreAK CRC-001 NCT06934538

Canonical source: Google.org GIC submission id. 53998912, filed 05/01/2026. All metrics, claims, and positioning statements trace to that document.

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